Supplementary Materials for "A Formal Investigation of Paradoxes Between Latent Aptitude and Observed Achievement via Full House Methodology"

This supplementary materials includes proofs of technical results and additional numerical examples supporting conclusions from Section 4 the main manuscript.

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1 Proofs

1.1 Proof of Theorem 2.2

Proof. Fix a population i and suppress the index i. First, since F_X is continuous, we can express well-defined order statistics as $X_{(1)} < \cdots < X_{(N)}$. By rank-based selection, the observed system consists of indices j such that $r_j \in \mathcal{R}$, yielding observed pairs $(Y_{(j)}, X_{(r_j)})$, for $j = 1, \ldots, n$. By assumption, for each observed rank j there exists a strictly increasing measurable function ψ_j such that

$$Y_{(j)} = \psi_j(X_{(r_j)}) \quad \text{a.s.}$$

We now identify the distribution function $F_{Y_{(j)}}$ for fixed j. Since ψ_j is strictly increasing, it admits a measurable inverse on its range. For any $y \in \mathbb{R}$,

$$\mathrm{P}(Y_{(j)} \leq y) = \mathrm{P}(\psi_j(X_{(r_j)}) \leq y) = \mathrm{P}(X_{(r_j)} \leq \psi_j^{-1}(y)).$$

Hence, $F_{Y_{(j)}}(y) = F_{X_{(r_j)}}(\psi_j^{-1}(y))$, which implies that

$$F_{Y_{(j)}}(\psi_j(x)) = F_{X_{(r_j)}}(x) \quad \text{for } F_{X_{(r_j)}}\text{-a.e. } x.$$

We now show that $Y_{(j)}$ is recovered from the Full House Modeling construction. Since F_X is continuous, the distribution of the order statistic $X_{(r_j)}$ is also continuous and strictly increasing on the interior of its support, hence admits a well-defined generalized inverse. Likewise, $F_{Y_{(j)}}$ is continuous and strictly increasing. Thus both admit well-defined inverses. Let $u = F_{X_{(r_j)}}(x)$. Then $F_{Y_{(j)}}(\psi_j(x)) = u$, so that

$$\psi_j(x) = F_{Y_{(j)}}^{-1}(u) = F_{Y_{(j)}}^{-1}\big(F_{X_{(r_j)}}(x)\big)$$

for $F_{X_{(r_j)}}$-almost every x. Therefore,

$$Y_{(j)} = F^{-1}_{Y_{(j)}} \circ F_{X_{(r_j)}}(X_{(r_j)}) \quad \text{a.s.}$$

We now verify the uniqueness of the above. Let $U = F_{X_{(r_j)}}(X_{(r_j)}) \sim \text{Unif}(0,1)$ by the probability integral transform, and define

$$g(u) := \psi_j\big(F^{-1}_{X_{(r_j)}}(u)\big).$$

Then g is strictly increasing and $g(U) \sim F_{Y_{(j)}}$. By the fact that if $U \sim \text{Unif}(0,1)$ and g is increasing, then the distribution of $g(U)$ uniquely determines g almost everywhere, implying $g(u) = F^{-1}_{Y_{(j)}}(u)$ a.e. It follows that

$$\psi_j(x) = F^{-1}_{Y_{(j)}}\big(F_{X_{(r_j)}}(x)\big) \quad \text{for } F_{X_{(r_j)}}\text{-a.e. } x.$$

Thus, the strictly increasing coupling is unique up to null sets. The representation holds for each observed rank j, and uniqueness implies that the joint distribution of (X, Y) coincides with that induced by the Full House Model. □

1.2 Proof of Lemma 3.2

Proof. Define $f_{1,j}(\cdot) = F^{-1}_{Y_{1,(j)}} \circ F_{X_{1,(r_{1,j})}}(\cdot)$ and $f_{2,k}(\cdot) = F^{-1}_{Y_{2,(k)}} \circ F_{X_{2,(r_{2,k})}}(\cdot)$. Then we have

$$\begin{aligned}
&\mathrm{P}\left(X_{1,(r_{1,j})} > X_{2,(r_{2,k})}, Y_{1,(j)} < Y_{2,(k)}\right) \\
&\quad = \mathrm{P}\left(X_{1,(r_{1,j})} > X_{2,(r_{2,k})}), f_{1,j}(X_{1,(r_{1,j})}) < f_{2,k}(X_{2,(r_{2,k})})\right) \\
&\quad = \mathrm{P}\left(X_{2,(r_{2,k})} < X_{1,(r_{1,j})} < h_{j,k}\left(X_{2,(r_{2,k})}\right)\right),
\end{aligned}$$

where $h_{j,k}(\cdot) = (f^{-1}_{1,j} \circ f_{2,k})(\cdot)$. A similar technique can be applied to the second equation to obtain the result. □

1.3 Proof of Theorem 3.4

Proof. By Lemma 3.2,

$$\mathrm{P}\left(X_{1,(r_{1,j})} > X_{2,(r_{2,k})}, Y_{1,(j)} < Y_{2,(k)}\right) = \mathrm{P}\left(X_{2,(r_{2,k})} < X_{1,(r_{1,j})} < h_{j,k}\left(X_{2,(r_{2,k})}\right)\right),$$

where $h_{j,k}(\cdot) = F^{-1}_{X_{1,(r_{1,j})}} \circ F_{Y_{1,(j)}} \circ F^{-1}_{Y_{2,(k)}} \circ F_{X_{2,(r_{2,k})}}(\cdot)$. Under the assumption $F_{Y_1} = F_{Y_2} = F_Y$, the distributions of the order statistics admit Beta representations, which allow us to express the mapping $h_{j,k}$ in terms of regularized incomplete beta functions.

$$\begin{aligned}
&\mathrm{P}\left(X_{2,(r_{2,k})} < X_{1,(r_{1,j})} < h_{j,k}\left(X_{2,(r_{2,k})}\right)\right) \\
&= \int_{-\infty}^{+\infty} \left\{ \int_{x}^{F_X^{-1} \circ I^{-1}_{r_{1,j},N_1-r_{1,j}+1} \circ I_{j,n_1-j+1} \circ I^{-1}_{k,n_2-k+1} \circ I_{r_{2,k},N_2-r_{2,k}+1} \circ F_X(x)} f_{X_{1,(r_{1,j})}}(y)\, dy \right\} \\
&\qquad \times f_{X_{2,(r_{2,k})}}(x)\, dx \\
&= \int_{-\infty}^{+\infty} \left\{ \int_{x}^{F_X^{-1} \circ I^{-1}_{r_{1,j},N_1-r_{1,j}+1} \circ I_{j,n_1-j+1} \circ I^{-1}_{k,n_2-k+1} \circ I_{r_{2,k},N_2-r_{2,k}+1} \circ F_X(x)} d\, I_{r_{1,j},N_1-r_{1,j}+1}(F_X(y)) \right\} \\
&\qquad \times f_{X_{2,(r_{2,k})}}(x)\, dx \\
&= \int_{-\infty}^{+\infty} \left\{ I_{j,n_1-j+1} \circ I^{-1}_{k,n_2-k+1} \circ I_{r_{2,k},N_2-r_{2,k}+1} \circ F_X(x) - I_{r_{1,j},N_1-r_{1,j}+1} \circ F_X(x) \right\}^+ \\
&\qquad \times f_{X_{2,(r_{2,k})}}(x)\, dx \\
&= \int_{-\infty}^{+\infty} \left\{ I_{j,n_1-j+1} \circ I^{-1}_{k,n_2-k+1} \circ I_{r_{2,k},N_2-r_{2,k}+1} \circ F_X(x) - I_{r_{1,j},N_1-r_{1,j}+1} \circ F_X(x) \right\}^+ \\
&\qquad d\, I_{r_{2,k},N_2-r_{2,k}+1}(F_X(x)) \\
&= \int_0^1 \left\{ I_{j,n_1-j+1} \circ I^{-1}_{k,n_2-k+1}(t) - I_{r_{1,j},N_1-r_{1,j}+1} \circ I^{-1}_{r_{2,k},N_2-r_{2,k}+1}(t) \right\}^+ dt.
\end{aligned}$$

This representation holds without additional assumptions. The positive-part operator appears because the integrand may change sign. Under condition (8), the integrand is nonneg-

ative, hence

$$\int_0^1 \left\{ I_{j,n_1-j+1} \circ I^{-1}_{k,n_2-k+1}(t) - I_{r_{1,j},N_1-r_{1,j}+1} \circ I^{-1}_{r_{2,k},N_2-r_{2,k}+1}(t) \right\}^+ dt$$
$$= \int_0^1 \left\{ I_{j,n_1-j+1} \circ I^{-1}_{k,n_2-k+1}(t) - I_{r_{1,j},N_1-r_{1,j}+1} \circ I^{-1}_{r_{2,k},N_2-r_{2,k}+1}(t) \right\} dt.$$

Thus,

$$\begin{aligned}
&\int_0^1 \left\{ I_{j,n_1-j+1} \circ I^{-1}_{k,n_2-k+1}(t) - I_{r_{1,j},N_1-r_{1,j}+1} \circ I^{-1}_{r_{2,k},N_2-r_{2,k}+1}(t) \right\}^+ dt \\
&\quad = \int_0^1 I_{j,n_1-j+1}(x)\, dI_{k,n_2-k+1}(x) - \int_0^1 I_{r_{1,j},N_1-r_{1,j}+1}(x)\, dI_{r_{2,k},N_2-r_{2,k}+1}(x) \\
&\quad = \int_0^1 I_{j,n_1-j+1}(x)\, \frac{x^{k-1}(1-x)^{n_2-k}}{B(k, n_2-k+1)}\, dx \\
&\qquad - \int_0^1 I_{r_{1,j},N_1-r_{1,j}+1}(x)\, \frac{x^{r_{2,k}-1}(1-x)^{N_2-r_{2,k}}}{B(r_{2,k}, N_2-r_{2,k}+1)}\, dx \\
&\quad = \int_0^1 \sum_{p=j}^{n_1} \binom{n_1}{p} x^p (1-x)^{n_1-p} \frac{x^{k-1}(1-x)^{n_2-k}}{B(k, n_2-k+1)}\, dx \\
&\qquad - \int_0^1 \sum_{p=r_{1,j}}^{N_1} \binom{N_1}{p} x^p (1-x)^{N_1-p} \frac{x^{r_{2,k}-1}(1-x)^{N_2-r_{2,k}}}{B(r_{2,k}, N_2-r_{2,k}+1)}\, dx \\
&\quad = \sum_{p=j}^{n_1} \binom{n_1}{p} \frac{B(k+p, n_2+n_1-p-k+1)}{B(k, n_2-k+1)} \\
&\qquad - \sum_{p=r_{1,j}}^{N_1} \binom{N_1}{p} \frac{B(r_{2,k}+p, N_2+N_1-p-r_{2,k}+1)}{B(r_{2,k}, N_2-r_{2,k}+1)}.
\end{aligned}$$

□

1.4 Proof of Proposition 3.5

Proof. Specification of the selection rule $S_{i,j} = 1 \iff X_{i,j} \geq X_{i,(N_i-n_i+1)}$ implies that $r_{i,j} = N_i + 1 - j$. When either $n_1 = N_1$ and $n_2 = N_2$ or $n_1 = n_2$ and $N_1 = N_2$ we have, for

all $t \in (0,1)$, that

$$I_{N_2-j,j+1} \circ I^{-1}_{N_1-j,j+1} \circ I_{n_1-j,j+1} \circ I^{-1}_{n_2-j,j+1}(t) \geq t.$$

The left hand side of the above equation increases whenever n_1 or N_1 are increased. This follows from the monotonicity of the inverse incomplete beta function in its parameters [Askitis, 2021], together with the fact that stochastic dominance is preserved under monotone transformations. Therefore, $I_{N_2-j,j+1} \circ I^{-1}_{N_1-j,j+1} \circ I_{n_1-j,j+1} \circ I^{-1}_{n_2-j,j+1}(t) \geq t$ whenever $\frac{n_1}{n_2} \leq 1$ and $\frac{N_2}{N_1} \leq 1$. Observe that

$$\begin{aligned}
&I_{N_2-j,j+1} \circ I^{-1}_{N_1-j,j+1} \circ I_{n_1-j,j+1} \circ I^{-1}_{n_2-j,j+1}(t) \geq t, \\
&\implies I_{n_1-j,j+1} \circ I^{-1}_{n_2-j,j+1}(t) \geq I_{N_1-j,j+1} \circ I^{-1}_{N_2-j,j+1}(t), \\
&\implies I_{n_1-j,j+1} \circ I^{-1}_{n_2-j,j+1}(x) - I_{N_1-j,j+1} \circ I^{-1}_{N_2-j,j+1}(t) \geq t.
\end{aligned}$$

Technical condition (8) holds and the conclusion of Theorem 3.4 applies. □

1.5 Proof of Proposition 4.2

Proof. First, in combination with the Definition 4.1 and the assumption $\frac{n_1}{N_1} \leq \frac{n_2}{N_2}$, we have

$$F_{Y_1}(x)^{\frac{n_1}{N_1}} \geq F_{Y_2}(x)^{\frac{n_2}{N_2}} \implies x \geq F^{-1}_{Y_1}\left(F_{Y_2}(x)^{\frac{n_2 N_1}{N_2 n_1}}\right).$$

By applying the Equation 7 from the Lemma 3.2 we have

$$\begin{aligned}
&\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) = \mathrm{P}\left(g_{n_1,n_2}\left(Y_{2,(n_2)}\right) < Y_{1,(n_1)} < Y_{2,(n_2)}\right) \\
&= \mathrm{P}\left(F^{-1}_{Y_1}\left(F_{Y_2}(Y_{2,(n_2)})^{\frac{n_2 N_1}{N_2 n_1}}\right) < Y_{1,(n_1)} < Y_{2,(n_2)}\right) \\
&= \int_{-\infty}^{+\infty} \int_{F^{-1}_{Y_1}\left(F_{Y_2}(x)^{\frac{n_2 N_1}{N_2 n_1}}\right)}^{x} dF_{Y_1,(n_1)}(y) dF_{Y_2,(n_2)}(x)
\end{aligned}$$

$$= \int_{-\infty}^{+\infty} F_{Y_1}(x)^{n_1} - F_{Y_2}(x)^{\frac{n_2 N_1}{N_2}} dF_{Y_{2,(n_2)}}(x)$$
$$= \int_0^1 F_{Y_1}\left(F_{Y_2}^{-1}(t^{\frac{1}{n_2}})\right)^{n_1} - t^{\frac{N_1}{N_2}} dt.$$

Based on the Definition 4.1, we have

$$\begin{aligned}
\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) &= \int_0^1 F_{Y_1}\left(F_{Y_2}^{-1}(t^{\frac{1}{n_2}})\right)^{n_1} - t^{\frac{N_1}{N_2}} dt \\
&\geq \int_0^1 F_{Y_2}\left(F_{Y_2}^{-1}(t^{\frac{1}{n_2}})\right)^{n_1} - t^{\frac{N_1}{N_2}} dt \\
&= \frac{n_2}{n_1+n_2} - \frac{N_2}{N_1+N_2},
\end{aligned}$$

and

$$\begin{aligned}
\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) &= \int_0^1 F_{Y_1}\left(F_{Y_2}^{-1}(t^{\frac{1}{n_2}})\right)^{n_1} - t^{\frac{N_1}{N_2}} dt \\
&\leq \int_0^1 \left(t^{1/n_2} + \varepsilon\right)^{n_1} - t^{\frac{N_1}{N_2}} dt \\
&= n_2 \sum_{i=0}^{n_1} \frac{\binom{n_1}{i}\varepsilon^i}{n_1+n_2-i} - \frac{N_2}{N_1+N_2}.
\end{aligned}$$

□

1.6 Proof of Theorem 4.4

Proof. Equation 7 from Lemma 3.2 yields

$$g_{n_1,n_2}(x) = F_{Y_1}^{-1}\left(F_{Y_2}(x)^{\frac{n_2 N_1}{N_2 n_1}}\right) = \Phi^{-1}\left(\Phi\left(\frac{x-\mu_2}{\sigma_2}\right)^{\frac{n_2 N_1}{N_2 n_1}}\right)\sigma_1 + \mu_1.$$

Let A and R be as defined, respectively, in (14) and (13). We have,

$$\begin{aligned}
\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) &= \mathrm{P}\left(g_{n_1,n_2}\left(Y_{2,(n_2)}\right) < Y_{1,(n_1)} < Y_{2,(n_2)}\right) \\
&= \mathrm{P}\left(F_{Y_1}^{-1}\left(F_{Y_2}(Y_{2,(n_2)})^{\frac{n_2 N_1}{N_2 n_1}}\right) < Y_{1,(n_1)} < Y_{2,(n_2)}\right)
\end{aligned}$$

$$
\begin{aligned}
&= \int_A \int_{F_{Y_1}^{-1}\left(F_{Y_2}(x)^{\frac{n_2 N_1}{N_2 n_1}}\right)}^{x} dF_{Y_1,(n_1)}(y) dF_{Y_2,(n_2)}(x) \\
&= \int_A \left[F_{Y_1}(x)^{n_1} - F_{Y_2}(x)^{\frac{n_2 N_1}{N_2}}\right] dF_{Y_2,(n_2)}(x) \\
&= \int_{-\infty}^{\infty} \left[F_{Y_1}(x)^{n_1} - F_{Y_2}(x)^{\frac{n_2 N_1}{N_2}}\right] dF_{Y_2,(n_2)}(x) - \int_{A^c} \left[F_{Y_1}(x)^{n_1} - F_{Y_2}(x)^{\frac{n_2 N_1}{N_2}}\right] dF_{Y_2,(n_2)}(x) \\
&= \int_{-\infty}^{\infty} \left[F_{Y_1}(x)^{n_1} - F_{Y_2}(x)^{\frac{n_2 N_1}{N_2}}\right] dF_{Y_2,(n_2)}(x) + R.
\end{aligned}
$$

We continue with the left-hand side of the last equality,

$$
\begin{aligned}
\int_{-\infty}^{\infty} \left[F_{Y_1}(x)^{n_1} - F_{Y_2}(x)^{\frac{n_2 N_1}{N_2}}\right] dF_{Y_2,(n_2)}(x) &= \int_0^1 \left[F_{Y_1}\big(F_{Y_2}^{-1}(t^{1/n_2})\big)^{n_1} - t^{N_1/N_2}\right] dt \\
&= \int_0^1 \left(\Phi\left(\frac{\Phi^{-1}(t^{1/n_2})\sigma_2 + \mu_2 - \mu_1}{\sigma_1}\right)\right)^{n_1} dt \; - \; \frac{N_2}{N_1+N_2} \\
&= \int_{-\infty}^{+\infty} \Phi\left(\frac{\sigma_2 x + \mu_2 - \mu_1}{\sigma_1}\right)^{n_1} \left(n_2 \Phi(x)^{n_2-1}\phi(x)\right) dx \; - \; \frac{N_2}{N_1+N_2} \\
&= n_2 \int_{-\infty}^{+\infty} \mathrm{P}\left(\begin{array}{l} Z_{1,1} - \frac{\sigma_2}{\sigma_1} X \le \frac{\mu_2-\mu_1}{\sigma_1}, \ldots, Z_{1,n_1} - \frac{\sigma_2}{\sigma_1} X \le \frac{\mu_2-\mu_1}{\sigma_1}, \\ Z_{2,1} - X \le 0, \ldots, Z_{2,n_2-1} - X \le 0 \end{array} \middle| \, X = x \right) \phi(x)\, dx \\
&\qquad - \; \frac{N_2}{N_1+N_2} \\
&= n_2 \, \mathrm{P}\left(\begin{array}{l} Z_{1,1} - \frac{\sigma_2}{\sigma_1} X \le \frac{\mu_2-\mu_1}{\sigma_1}, \ldots, Z_{1,n_1} - \frac{\sigma_2}{\sigma_1} X \le \frac{\mu_2-\mu_1}{\sigma_1}, \\ Z_{2,1} - X \le 0, \ldots, Z_{2,n_2-1} - X \le 0 \end{array} \right) \; - \; \frac{N_2}{N_1+N_2} \\
&= n_2 \, F_{\mathbf{W}}(\mathbf{w}) \; - \; \frac{N_2}{N_1+N_2},
\end{aligned}
$$

where the first n_1 elements of $\mathbf{w}$ is $\frac{\mu_2-\mu_1}{\sigma_1}$ and the last $n_2 - 1$ elements are 0. The distribution function $F_{\mathbf{W}}$ corresponds to random variable $\mathbf{W} \sim \mathcal{N}_{(n_1+n_2-1)}(\mathbf{0}, \mathbf{\Sigma})$ where $\mathbf{\Sigma}$ is as defined in (16) and (17).

The bound on the remainder (15) follows from the argument in Section 4.2. □

1.7 Proof of Theorem 4.5

Proof. From (18) we have

$$\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) = n_2 F_{\mathbf{W}}(\mathbf{w}) - \frac{N_2}{N_1+N_2} + R,$$

and we estimate the above with

$$\widehat{\mathrm{P}}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) = n_2 \widehat{F}_{\mathbf{W}}(\mathbf{w}) - \frac{N_2}{N_1+N_2} + \widehat{R},$$

where $\widehat{F}_{\mathbf{W}}(\mathbf{w})$ and $\widehat{R}$ are obtained by replacing μ_i, σ_i with estimators $\hat{\mu}_i, \hat{\sigma}_i$, $i = 1, 2$.

By Theorem 4.4, $|R| \leq \left(\frac{1}{2}\right)^{n_2}$ under the conditions $\frac{n_1}{N_1} \leq \frac{n_2}{N_2}$, $\sigma_1 \leq \sigma_2$, and $\mu_1 \leq \mu_2$. These conditions are satisfied in our setting. Let $E = \{\hat{\sigma}_1 > \hat{\sigma}_2\} \cup \{\hat{\mu}_1 > \hat{\mu}_2\}$. Then $|\widehat{R}| \leq \left(\frac{1}{2}\right)^{n_2}$ on E^c. Our convergence of probability conditions guarantee that $\mathrm{P}(E^c) \to 1$ as $n_1, n_2 \to \infty$.

Now we show convergence of $\widehat{F}_{\mathbf{W}}(\mathbf{w})$. From the last equality in the proof of Theorem 4.4 we can write

$$\begin{aligned} F_{\mathbf{W}}(\mathbf{w}) &= \mathrm{P}\left(\begin{array}{l} Z_1 + \frac{\sigma_2}{\sigma_1} Z_0 \leq \frac{\mu_2-\mu_1}{\sigma_1}, \ldots, Z_{n_1} + \frac{\sigma_2}{\sigma_1} Z_0 \leq \frac{\mu_2-\mu_1}{\sigma_1}, \\ Z_{n_1+1} + Z_0 \leq 0, \ldots, Z_{n_1+n_2-1} + Z_0 \leq 0 \end{array} \right) \\ &= \mathrm{P}\left(\begin{array}{l} Z_1 \leq \frac{\mu_2-\mu_1}{\sigma_1} - \frac{\sigma_2}{\sigma_1} Z_0, \ldots, Z_{n_1} \leq \frac{\mu_2-\mu_1}{\sigma_1} - \frac{\sigma_2}{\sigma_1} Z_0, \\ Z_{n_1+1} \leq -Z_0, \ldots, Z_{n_1+n_2-1} \leq -Z_0 \end{array} \right) \end{aligned}$$

where $Z_i \overset{iid}{\sim} N(0,1)$, $i = 0, \ldots, n_1 + n_2 + 1$. Define $M_{n_1} = \max\{Z_1, Z_2, \ldots, Z_{n_1}\}$, $M_{n_2-1} = \max\{Z_{n_1+1}, Z_{n_1+2}, \ldots, Z_{n_1+n_2-1}\}$, then we can write

$$F_{\mathbf{W}}(\mathbf{w}) = \mathrm{P}\left(M_{n_1} \leq \frac{\mu_2-\mu_1}{\sigma_1} - \frac{\sigma_2}{\sigma_1} Z_0, \;\; M_{n_2-1} \leq -Z_0 \right).$$

From Embrechts et al. [2013], we have $M_{n_1} = \sqrt{2 \ln n_1} + O_P(1)$. Thus, after rearranging, the

inequality $M_{n_1} \leq \frac{\mu_2-\mu_1}{\sigma_1}$ is approximately equivalent (up to $O_P(1)$ fluctuations) to $\sqrt{2\ln n_1} \leq \frac{\mu_2-\mu_1}{\sigma_1} - \frac{\sigma_2}{\sigma_1} Z_0$. Rearranging yields

$$Z_0 \leq \frac{\mu_2-\mu_1}{\sigma_2} - \frac{\sigma_1}{\sigma_2}\sqrt{2\ln n_1}.$$

Similarly, from Embrechts et al. [2013], $M_{n_2-1} = \sqrt{2\ln(n_2-1)} + O_P(1)$. Thus, after rearranging, the inequality $M_{n_2-1} \leq -Z_0$ is approximately equivalent to $Z_0 \leq -\sqrt{2\ln(n_2-1)}$. Then, as $n_1, n_2 \to \infty$, we have

$$F_{\mathbf{W}}(\mathbf{w}) = \mathrm{P}\left(Z_0 \leq \frac{\mu_2-\mu_1}{\sigma_2} - \frac{\sigma_1}{\sigma_2}\sqrt{2\ln n_1},\ \ Z_0 \leq -\sqrt{2\ln(n_2-1)}\right).$$

Define

$$C := \frac{\mu_2-\mu_1}{\sigma_2} - \frac{\sigma_1}{\sigma_2}\sqrt{2\log n_1}, \qquad B := -\sqrt{2\log(n_2-1)},$$

and similarly

$$\widehat{C} = \frac{\hat{\mu}_2-\hat{\mu}_1}{\hat{\sigma}_2} - \frac{\hat{\sigma}_1}{\hat{\sigma}_2}\sqrt{2\log n_1}.$$

Two asymptotic regimes arise depending on whether $C \leq B$ or $C \geq B$. Under the consistency conditions [need to define and then argue], the plug-in estimator selects the same regime with probability tending to one.

First, consider the asymptotic regime $C \leq B$ where

$$F_{\mathbf{W}}(\mathbf{w}) = \mathrm{P}(Z_0 \leq C)\big(1+o(1)\big) = \Phi(C)\big(1+o(1)\big).$$

Since $C \to -\infty$, Mills' ratio $\Phi(-x) \sim \phi(x)/x$ gives

$$\Phi(C) \ \sim\ \frac{1}{\sqrt{2\pi}\,|C|}\ \exp\left(-\tfrac{1}{2}\left(\tfrac{\mu_2-\mu_1}{\sigma_2} - \tfrac{\sigma_1}{\sigma_2}\sqrt{2\ln n_1}\right)^2\right).$$

Let $\widehat{F}_{\mathbf{W}}(\mathbf{w})$ be obtained by replacing μ_i, σ_i with consistent estimators $\hat{\mu}_i, \hat{\sigma}_i$, and define

$$\widehat{C} := \frac{\hat{\mu}_2 - \hat{\mu}_1}{\hat{\sigma}_2} - \frac{\hat{\sigma}_1}{\hat{\sigma}_2}\sqrt{2\ln n_1}.$$

Replacing C with $\widehat{C}$ yields

$$\widehat{F}_{\mathbf{W}}(\mathbf{w}) \;\sim\; \frac{1}{\sqrt{2\pi}\,|\widehat{C}|}\,\exp\!\left(-\tfrac{1}{2}\left(\tfrac{\hat{\mu}_2-\hat{\mu}_1}{\hat{\sigma}_2} - \tfrac{\hat{\sigma}_1}{\hat{\sigma}_2}\sqrt{2\ln n_1}\right)^2\right),$$

and therefore

$$\frac{\widehat{F}_{\mathbf{W}}(\mathbf{w})}{F_{\mathbf{W}}(\mathbf{w})} \;\sim\; \frac{|C|}{|\widehat{C}|}\,\exp\!\left(-\tfrac{1}{2}\big(\widehat{C}^{\,2} - C^2\big)\right).$$

Writing out the squares,

$$\widehat{C}^{\,2} - C^2 = \underbrace{\left[\left(\tfrac{\hat{\mu}_2-\hat{\mu}_1}{\hat{\sigma}_2}\right)^2 - \left(\tfrac{\mu_2-\mu_1}{\sigma_2}\right)^2\right]}_{o_P(1)} + (2\ln n_1)\left[\frac{\hat{\sigma}_1^{\,2}}{\hat{\sigma}_2^{\,2}} - \frac{\sigma_1^{\,2}}{\sigma_2^{\,2}}\right] - 2\sqrt{2\ln n_1}\left[\frac{\hat{\mu}_2-\hat{\mu}_1}{\hat{\sigma}_2}\cdot\frac{\hat{\sigma}_1}{\hat{\sigma}_2} - \frac{\mu_2-\mu_1}{\sigma_2}\cdot\frac{\sigma_1}{\sigma_2}\right].$$

The first bracket is $o_P(1)$ under any consistent estimator. The remaining two terms grow with n_1 at rates $\ln n_1$ and $\sqrt{\ln n_1}$, so they govern the asymptotic behavior of $\frac{\widehat{F}_{\mathbf{W}}(\mathbf{w})}{F_{\mathbf{W}}(\mathbf{w})}$.

Provided $\hat{\sigma}_1/\hat{\sigma}_2 \xrightarrow{P} \sigma_1/\sigma_2$, the prefactor $|C|/|\widehat{C}|$ tends to 1 in probability. Combined with the exponent expansion,

$$\frac{\widehat{F}_{\mathbf{W}}(\mathbf{w})}{F_{\mathbf{W}}(\mathbf{w})} \xrightarrow{P} 1$$

whenever the assumed conditions,

$$(2\ln n_1)\left|\frac{\hat{\sigma}_1^{\,2}}{\hat{\sigma}_2^{\,2}} - \frac{\sigma_1^{\,2}}{\sigma_2^{\,2}}\right| \xrightarrow{P} 0,$$

$$\sqrt{2\ln n_1}\left|\frac{\hat{\mu}_2-\hat{\mu}_1}{\hat{\sigma}_2}\cdot\frac{\hat{\sigma}_1}{\hat{\sigma}_2} - \frac{\mu_2-\mu_1}{\sigma_2}\cdot\frac{\sigma_1}{\sigma_2}\right| \xrightarrow{P} 0,$$

are satisfied.

Now consider the regime $B \leq C$. In this case we have $F_{\mathbf{W}}(\mathbf{w}) = \Phi(B)(1 + o(1))$. Since $B = -\sqrt{2\log(n_2 - 1)}$, Mills' ratio yields

$$F_{\mathbf{W}}(\mathbf{w}) \sim \frac{1}{\sqrt{4\pi \log(n_2 - 1)}} \cdot \frac{1}{n_2 - 1}.$$

Under consistency of the plug-in estimators, $\widehat{C} - C \xrightarrow{P} 0$, so whenever the inequality $B < C$ holds asymptotically with nonvanishing separation, the plug-in estimator selects the same regime with probability tending to one. Consequently, $\widehat{F}_{\mathbf{W}}(\mathbf{w}) = \Phi(B)(1+o_P(1))$, and hence

$$\frac{\widehat{F}_{\mathbf{W}}(\mathbf{w})}{F_{\mathbf{W}}(\mathbf{w})} \xrightarrow{P} 1.$$

Thus,

$$\begin{aligned}
&\widehat{\mathrm{P}}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) - \mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) \\
&\quad = n_2 \widehat{F}_{\mathbf{W}}(\mathbf{w}) - \frac{N_2}{N_1 + N_2} + \widehat{R} - \left(n_2 F_{\mathbf{W}}(\mathbf{w}) - \frac{N_2}{N_1 + N_2} + R\right) \\
&\quad = n_2 \left(\widehat{F}_{\mathbf{W}}(\mathbf{w}) - F_{\mathbf{W}}(\mathbf{w})\right) + \left(\widehat{R} - R\right) \xrightarrow{P} 0,
\end{aligned}$$

noting that $\widehat{F}_{\mathbf{W}}(\mathbf{w}) = o_P\left(F_{\mathbf{W}}(\mathbf{w})\right)$ and $n_2 F_{\mathbf{W}}(\mathbf{w}) = O_P(1)$. □

2 Simulations

2.1 Simulation Results for Theorem 4.4

We now show the simulation results of our Normal Distribution example under different configurations of $(\mu_1, \mu_2, \sigma_1, \sigma_2, n_1, n_2, N_1, N_2)$. We verify Theorem 4.4 in the numerical simulation. Directly paradox probabilities from these theorems are compared with Monte Carlo paradox probabilities in a variety of settings. Each of these settings will involve comparisons of the observed statistics and latent aptitudes of the highest achievers, the pair

$(Y_{i,(n_i)}, X_{i,(N_i)})$ across two populations.

The simulations will proceed as follows: Distributions for Y_1 and Y_2 are specified along with sample sizes n_i, N_i, $i = 1, 2$. We will specify a Monte Carlo sample size of 2000. For iteration m of the Monte Carlo sample procedure $Y_{1,j}$, $j = 1, \ldots, n_1$ and $Y_{2,k}$, $k = 1, \ldots, n_2$ are generated. Then $X_{i,(N_i)}$, $i = 1, 2$ is calculated using (2) where $X \sim \text{Pareto}(\alpha = 1.16)$. Then $1(X_{(N_1)} > X_{(N_2)}, Y_{(n_1)} < Y_{(n_2)})$ is recorded.

The theoretical results are computed from the Theorem 4.4 using mvtnorm package [Genz and Bretz, 2009] in R.

The results are tabulated in Table 1. These results show strong agreement between our theory and numerical simulation. They also reveal that paradox probabilities are non-trivial across these configurations.

n_1	n_2	N_1	N_2	μ_1	μ_2	σ_1	σ_2	Simulation Prob	Theoretical Prob
10	40	400	600	-1	1	1.0	3	0.4000	0.4000099
10	20	400	600	0	0	1.0	3	0.3970	0.3970305
10	40	400	300	-1	1	1.0	3	0.5715	0.5714174
10	20	400	600	0	1	1.5	2	0.3645	0.3647783
10	40	400	600	0	1	1.0	3	0.3995	0.3999245
20	40	400	600	0	1	1.0	3	0.3995	0.3999803
10	40	400	600	-1	0	1.5	3	0.4000	0.3995149
20	40	400	600	0	0	1.5	2	0.2970	0.2962431
20	40	400	600	0	0	1.0	3	0.3750	0.3740938
10	20	400	600	-1	1	1.0	3	0.3990	0.4000059

Table 1: Simulation Results for Normal Example.

References

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